

Semantic Priming in Word Clouds

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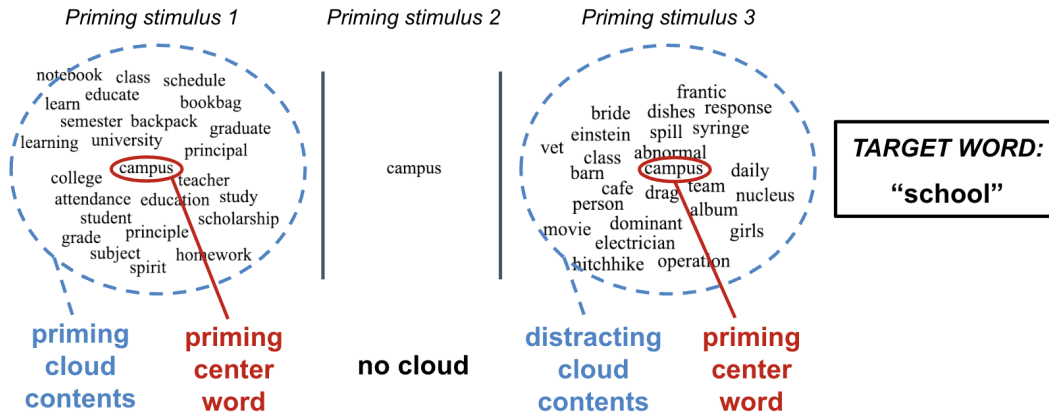


Figure 1: Types of stimuli: The leftmost stimulus uses a cloud to prime the reader for the word “school.” The middle stimulus has no cloud but still primes for “school.” The right stimulus has a distractor cloud but the center word still primes for “school.”

ABSTRACT

While word clouds pack a lot of data into a relatively small area, it is unclear whether readers actually benefit from all of that information, or if they are only processing a few of the words shown. We sought to determine if words outside of a reader’s central vision were contributing to their interpretation by leveraging the *semantic priming effect*: a phenomenon in which participants will more quickly recognize a word if they have been recently “primed” with a word that is semantically related. We presented participants with word clouds containing related and unrelated words to see whether they might prime this lexical decision task more strongly than single words alone. We showed that the peripheral contents of a cloud do affect participant performance at this task, though more work is needed to understand the impacts of these differences in real-world settings.

Keywords: word clouds, semantic priming, text perception

Index Terms: Human-centered computing—Visualization—Empirical studies in visualization;

1 INTRODUCTION

Word clouds are among the most common means of summarizing text data, albeit not without controversy. Despite their ubiquity in social media, newspaper articles, and some academic settings, many have questioned the accuracy with which they present information or have highlighted their potential for bias. Some have tried to measure and potentially correct for such drawbacks—e.g., identifying how a word’s shape might bias the reading of its associated data [1], or improving the *positions* of words in a cloud to better highlight their semantic groupings [9]. In this paper, we address a different potential drawback: **do readers actually gain anything from the**

words in a cloud other than the few in the center? Word clouds are quite space efficient, managing to convey information with no additional marks—nothing but data ink! However, this benefit only really matters if people are actually gaining insight from all of the words. If, as some have suggested [10], readers are just comprehending a *few* of the words (e.g., the large and/or central ones), then having extra information packed into the periphery might be futile.

We were curious if some benefit might be gained from additional words even if they aren’t actively attended to. In other words, is it possible that non-central words might strengthen or add nuance to a reader’s impression or gist of a word cloud regardless of whether the reader actively reads through each one of them?

To answer this question, we leveraged **semantic priming**, an effect by which one can more readily recognize a word if they have been “primed” by having just had a word from a related topic flashed in front of their vision. This phenomenon has been frequently replicated [13], though it has been unclear whether it operates outside a person’s central (foveal) vision, or whether it can be strengthened by *multiple* “priming” stimuli. In our experiment, we recreated a classical priming study, but showed participants word clouds rather than single words. If an entire *cloud* full of priming words create a stronger effect than a single word and/or a cloud of *distracting* (i.e., not semantically related) words, then it would seem to indicate that some semantic information about the words in the cloud was influencing the reader’s conception of the data being shown.

We ultimately found that the contents of a surrounding word cloud *do* have a measurable effect on semantic priming, indicating that the meanings of non-central words likely do contribute to a reader’s interpretation of the data. It is not yet clear whether this effect comes from priming words strengthening the priming effect or distracting words detracting from it, or both. This invites further study into this phenomenon.

2 BACKGROUND

2.1 Word clouds as a visualization tool

Word clouds are often used in summary and textual analysis applications [11] and have maintained growing popularity despite several shortcomings, such as their inability to provide an accurate summary

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of a text and ineffective word placement [9, 20]. Some studies have shown that word clouds are at least moderately useful for the purpose of gist forming [2, 9]. However, word clouds are less useful for other tasks such as summary [9], search, and comparison [6].

Perceptual and cognitive biases may limit word clouds' effectiveness. For example, people tend to focus on a specific part of the cloud, such as the top left [20]. Often, location (centrality) and font size form redundant encodings, with larger words preferentially placed toward the center, potentially *overemphasizing* those words relative to their associated data. Because of these existing biases, it is worthwhile to investigate whether users attend to or derive meaning from the other words in the cloud, such as those in outer areas.

2.2 Perceptual research into the priming mechanism

Semantic priming is typically investigated using the **lexical decision task**, where participants are shown a word (either prime or distractor) and then asked if an ensuing string of characters (called the "target") is a word. Participants typically identify target words faster after being shown related words (primes) (e.g., rain-UMBRELLA) than after being shown unrelated words (distractors) (e.g., shoes-UMBRELLA) [3, 13]. Evidence suggests that varying prime presentation times ranging from 80ms to 250ms still reliably produce the semantic priming effect [7, 12]. In addition, when the prime word is masked through transpositions (e.g. judge-COURT), the meaning of a prime word can still be processed and lead to a benefit in response time in word recognition [19].

Although in a word cloud, multiple words are presented simultaneously, readers attend to individual words sequentially. We sought to determine whether there might be some semantic information gleaned from words to which a reader doesn't actively attend, especially given that words in a word cloud are often semantically related. As such, we decided to use the semantic priming phenomenon and the lexical decision task as a potential metric for investigating the gist-forming effects of non-central words in a word cloud.

2.3 Bring it all together

In employing the priming effect in questions of word cloud perception, we hope to learn something about both word clouds and the semantic priming effect itself. For instance, there is contradictory evidence on whether words in one's non-central vision can influence viewers [4, 12, 18]. Similarly, there is evidence that not all words in a cloud are attended to equally [1, 20], leaving some parts of the cloud potentially out of central fixation. The priming effect gives us a unique way of testing this by allowing us to control for semantic relation between different parts of the cloud and the target word.

In perceptual research, most classical experiments concerning semantic priming present a single prime word and a single target. However, when participants actively look at certain words in a cloud, the rest of the words naturally lie in their non-central or peripheral vision. In that sense, using word clouds as stimuli produces two benefits. Firstly, it gives us insight on whether multiple related words lead to a stronger priming effect than single ones. If we find this to be the case, this also provides mechanisms for why people perform gist-forming tasks better when words are in semantic groups [9].

We leverage and test the priming effect to examine the degree of perception afforded by word clouds. Our results are of interest to researchers in both data visualization and perceptual psychology as they offer insight into the priming effects of multiple words as well as into the effectiveness of word clouds as data visualizations.

3 EXPERIMENTAL DESIGN

Given our goal of determining whether a reader is able to gain any semantic understanding from words to which they do not actively attend, we designed an experiment around the *semantic priming effect* described in previous sections. We sought to test whether a cloud of priming words would exhibit a stronger priming effect than

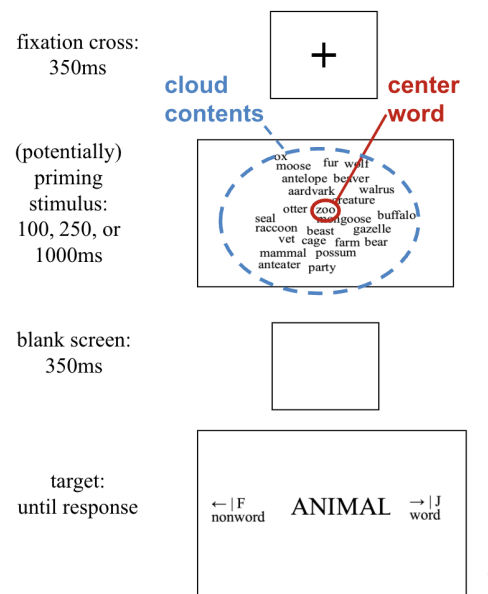


Figure 2: An example 4-stage trial. Participants first encountered a blank screen with a cross to fixate on in the center for 350ms before being shown the (potentially) priming stimulus. After a set amount of time, they were shown another blank screen for 350ms followed by the target word of the lexical decision task. We expected that if the priming stimulus was semantically related to the target word, participants would respond faster.

a single priming word, even if the participant did not have time to actively saccade to multiple words. To test this effect, we presented participants with a series of four-step trials (illustrated in Figure 2). First, they were shown a cross on which to fixate their gaze; next, they were presented with a **"priming" stimulus** (the contents of which varied across factors); then they were briefly shown a blank screen; finally, they were shown a **"target word."** Their task was to determine whether the presented target was a real English word, or a non-word (generated to be formed of pronounceable syllables, but not appearing in an English dictionary). As described in Section 2.2, response time on this lexical decision task decreases when the priming stimulus is semantically related to the target word.

The (potentially) priming stimuli with which we presented participants were broken down into the following factors and levels:

- **Center word:** The word at the center of the stimulus, placed directly in the same position as the fixation cross to which the participant attended. This would be either **priming** (i.e., semantically related to the target word) or a **distractor** (i.e., an English word that was *not* related to the target word).
- **Cloud content:** In some stimuli, we surrounded the center word with a collection of other words, resembling a word cloud (albeit a relatively bland one). The contents of these clouds were either **priming words**, **distractor words**, or simply **blank** (as a control comparison).
- **Display time:** The priming stimulus was presented for either a short time: **100ms**, or a long time: **250ms or 1000ms**. 250ms and 1000ms are grouped together in our analysis because they are longer than the time it takes to saccade (200ms) [8], below which the priming effect worsens for lexical decision tasks [15].

3.1 Hypotheses

It is worth noting that due to the structure of the lexical decision task, we are only really considering the evidence from the trials where the target is, in fact, a word (as only then can the preceding stimulus be potentially “priming”). We also hypothesized that we would not see any effects on *accuracy* within this task, which is typically near perfect and thus creates ceiling effects. For participant response time, we had the following hypotheses:

H1 (Center word): Response time will be shorter when the center word is priming compared to when it is not priming. This hypothesis is used to verify that our experiment effectively replicates classical priming studies [13] that use one priming word and one target word (identical to our no-cloud, priming condition).

H2 (Cloud contents): Response time will be the lowest when the center word is surrounded by a cloud of priming words, followed by the condition when there is no surrounding cloud, and highest for the distractor cloud.

H3 (Display time): Priming effects of the surrounding cloud (as articulated in H2) will be stronger when the display time is longer.

3.2 Stimuli

The word clouds in our stimuli were created using [17], positioned using a spiral layout. In order to isolate the effect of centrality, our clouds are all the same font size, color, word orientation (horizontal), and relative shape (circular). While this amount of control comes with a cost of ecological validity, it better helps us determine the *existence* of a priming effect associated with non-central words (even if the *size* of that effect within more realistic word clouds would need to be investigated as part of future work).

Each target was a string of Latin characters that made up either an English word or a nonsense word. English words were filtered to be easily recognizable to a native speaker without being stopwords (with a frequency between 3 and 500 appearances per million words in the SUBTLWF-US directory [5]). The nonsense words were all pronounceable nonwords with lengths between 3 and 8 letters¹.

The prime-target relationships were determined using the University of South Florida (USF) Free Association Norms [16] at a forward strength of 0.025 or greater, meaning the proportion of participants who said the target word when prompted only with the priming word was at least 2.5%. This association gives us reasonable confidence that our priming words do, in fact, prime participants for our target words. The chosen words were used in different factors of the stimuli as described in section 3. For stimuli with non-empty cloud content, clouds contained 16 to 25 words, some generated with word2vec² [14]. The source and quantity of words showed no significant effects in our results.

3.3 Experiment

The experiment followed a within-subjects design and consisted of 120 trials, each following the sequence of Figure 2: 6 trials per each of the 18 combinations of our factor levels (based off of a power analysis on H2), along with 6 “burn-in” trials immediately following the task instructions and again after a break halfway through the experiment. (In pilot experiments, we looked for potential learning effects, but saw performance level off quickly—hence the burn-in stimuli. We saw no learning effects in the final results.)

3.4 Participants

Based off a power analysis of pilot data seeking statistical power of 0.8 for a main effect on **H2** (see Section 3.1), we recruited 150

participants from Amazon Mechanical Turk. Each was paid between \$2.00 and \$2.50 for their time, typically 5-10 minutes for the whole experiment. Participants were native English speakers between the ages of 27 and 65, 46% female and 54% male.

4 RESULTS

As expected, we saw no effects on accuracy, which was 99% overall, with no conditions dipping below 97.4%. For Hypotheses 1-3, we ran within-subjects, one-way analyses of variance (ANOVAs) and used Holm’s method for *p* adjustment for pairwise tests between levels when appropriate. We excluded 3 data points (out of 8100) where participant’s response time was higher than 8 seconds (where the participant seems to have been distracted or stepped away from the task). The top row of Figure 3 shows the main effects on response time for the factors of center word, cloud contents, and display time, respectively.

Regarding **H1**, we observed a main effect of whether or not the center word in the stimulus was a priming word ($F(1, 8100) = 6.22, p = .013$). Participants were on average 18 ms faster when the center word was priming. This replicates the robust phenomenon from prior work, lending validation to our experiment design.

Regarding **H2**, we observed a main effect of the content of the surrounding cloud within the stimuli ($F(1, 8100) = 9.27, p = .002$). Post-hoc pairwise comparisons show a significant difference ($p = .035$) between response times when cloud contents are distractors ($M = 761ms, SD = 321$) and when cloud contents are primes ($M = 738ms, SD = 329$). However, we did not find a significant difference between the no-cloud condition ($M = 747ms, SD = 352$) and either of the two cloud conditions. Breaking these down further by the center word factor, we see that an effect of cloud contents exists when the center word is a priming word ($F(1, 3926) = 9.74, p = .002$) but not when the center word is a distractor ($F(1, 3923) = 1.56, p = .21$). In both conditions, the trend of response times is in the direction of our hypotheses (Figure 3-A2).

For **H3**, when display time is short (100 ms), we saw only marginal significance in the main effect of cloud contents ($F(1, 2565) = 3.42, p = .064$), as shown in Figure 3-C2. However, when display time was longer (250ms or 1000ms), the main effect of cloud contents was significant at the $\alpha = 0.05$ level ($F(1, 5284) = 5.85, p = .016$). There is also a main effect of display time (Figure 3-C1) ($F(1, 8100) = 4.346, p = .037$).

5 DISCUSSION

Ultimately, we were able to conclude that participants were indeed affected by non-central words: compared to distractors, priming words in the surrounding cloud decreased response time for the lexical decision task. It is worth noting that it is unclear whether the distractors “harm” recognition or the priming words “facilitate” recognition (or both), since comparison to the no-cloud group was not significant. Either way, we see that additional words do influence lexical decision. Though further analysis only showed significance for this effect when the center word was priming (Figure 3-A2), this may be due to a “floor effect,” in which the word recognition task can only be so slow, even when hampered by distraction—effectively capping the pink distracting-center/distracting-cloud bar.

Unfortunately, seeing only marginal significance for the main effect of cloud contents under the 100ms condition keeps us from being completely sure whether or not participants are gaining benefit from saccading to multiple words or relying on their parafoveal vision for the increased priming. It is possible that we are seeing a limit of the statistical power of the short display time condition, as it had half as many stimuli as the long display time condition (containing both 250ms and 1000ms). We hope to investigate this further with use of an eye tracker.

The main effect of time can be potentially attributed to participants having more time to see and process words, but it is puzzling

¹Generated from <https://www.soybomb.com/tricks/words/>

²Clouds for targets with fewer than 16 related words in the USF dataset (≥ 0.025 forward association strength) were supplemented using word2vec. The variance between 16 and 25 was due to the lack of strong matches from both the USF study and word2vec.

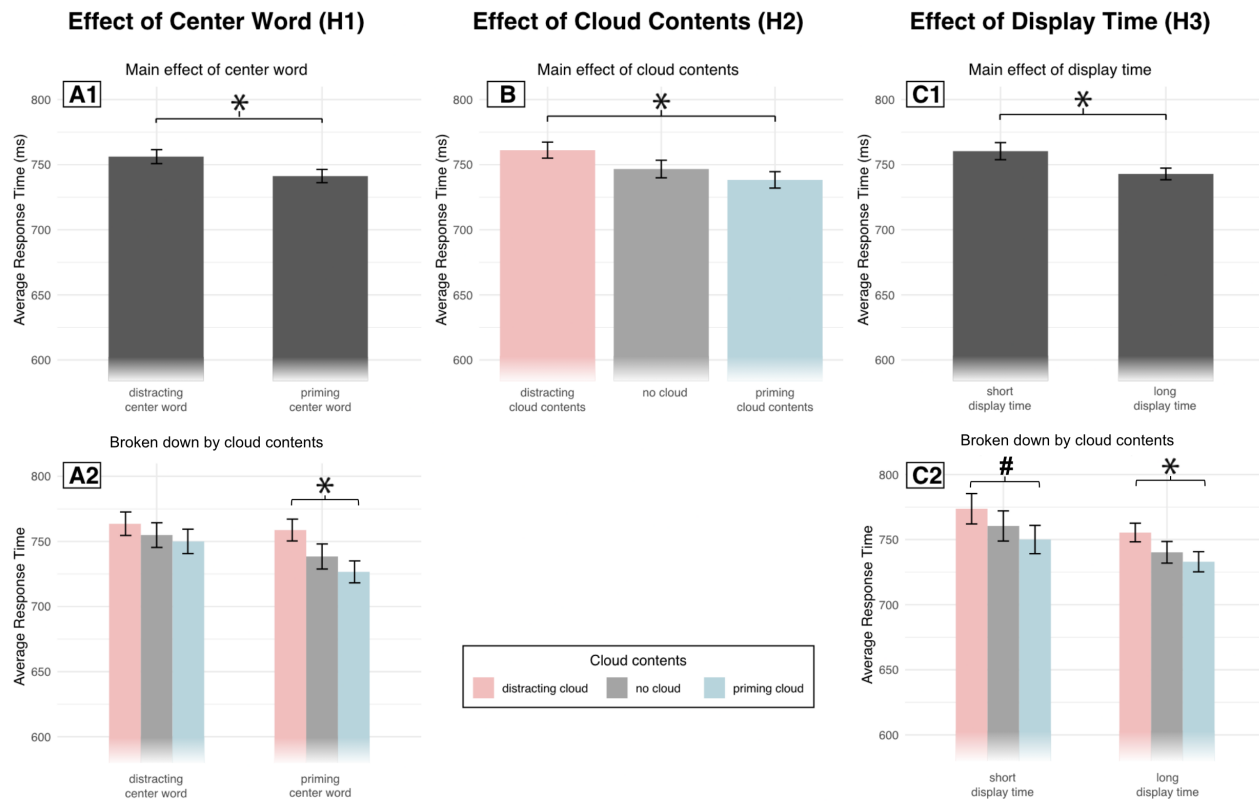


Figure 3: This figure shows main effects observed for the three primary factors of the study (top row), along with the effects broken down by whether the cloud shown contains priming or distracting words (bottom row). Error bars indicate standard deviation, while asterisks (*) indicate main effects that are significant at the $\alpha = 0.05$ level. Pound signs (#) indicate marginal significance at the $\alpha = 0.07$ level.

that extra time to peruse distracting clouds might be helpful in addition to extra time perusing priming clouds. Perhaps longer display time helps participants better prepare for the lexical decision task.

Regarding the implications of these results for word cloud visualizations, the main effect seen in Figure 3-B seems to show that people gain semantic information from surrounding words, contrary to prior concerns that there is no value to any but a handful of words in a cloud. This could be an indication that the high two-dimensional density of word clouds has the potential to enhance cognitive tasks such as gist-forming. The lack of pairwise significance with the “no cloud” condition makes it difficult to say whether we are seeing such enhancement in the data or merely the harmful effects of distracting words. We may be running into an issue of an insufficient sample. (We chose our sample size based on a power analysis performed on pilot data, seeking statistical power of 0.8 for a *main* effect on **H2**, which we did observe.) Nonetheless, the consistent trends we observe when breaking data down by cloud contents (Figures 3-A2, 3-B, and 3-C2), even if they do not reach full significance, suggest that future investigation into this effect could be fruitful.

While our results do not definitively show that non-central words in a word cloud contribute *positively* to a reader’s gist-forming abilities, they indicate that such contribution may be happening, and suggest some future directions. First, while we use a fixation cross and limit display time to try controlling where participants look, the use of an eye tracker could better inform which words are actually outside participants’ foveal vision. Second, our stimuli are much simpler than word clouds generally found in the wild. Investigating the priming effects of clouds containing varying font sizes, colors,

or other visual attributes could tell us more about how these results generalize. Finally, semantic priming is at best a proxy for the influence that non-central words might have on a reader’s impression of the data in a word cloud. It would be worthwhile to investigate the effects of non-central words on more real-world *tasks* as well as more realistically varied stimuli, though the often subjective nature of such tasks makes it difficult to do so in a controlled manner.

6 CONCLUSION

We sought to determine whether readers gain semantic information from words in a word cloud to which they might not actively attend. We leveraged *semantic priming*, an effect by which a person can more readily recognize a word if they have been “primed” by a related word immediately prior. We tested to see whether this effect was stronger when the priming stimulus was not just a single word, but an entire cloud of related words. We saw an effect of cloud contents, indicating that the priming effect was stronger when a cloud contained related words rather than distracting words. This suggests that readers to gain semantic information from the broader set of words. However, more study is needed to determine whether the addition of words in a cloud truly help with comprehension, or if unrelated words in a cloud are merely a hindrance.

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